

Punctuation

APPOSITIVES

USAGE

The only true appositive is one that is exclusively interchangeable. In that case, it is set off by commas. Otherwise, the person or thing being referred to is not set off by commas. (The correct version is marked by an asterisk.)

*A famous geographer, Henry Smith, is reported to have...
A famous geographer Henry Smith is reported to have...

*The best of these series, *Studies in Human Understanding*, gives...
The recent series, *Studies in Human Understanding*, gives...

APOSTROPHES

USAGE

1) The **possessive case of a singular noun** is formed by the addition of an apostrophe and an “s,” even if the noun ends in “s” or another sibilant. Of course, the editor has the option of turning “leaf’s color” into “the color of the leaf.”

person’s, deer’s, ibis’s, fox’s, one’s

Exceptions

Chicago Manual of Style makes exceptions for tradition and euphony:
for appearance’ (conscience’, righteousness’) sake

2) Possessive case of proper nouns

Burns’s poems, Marx’s theories, Nicholas’s death, Dickens’s novels
plural: the Rosses’ and the Williamses’ land (but see 3 below)

Exceptions

Jesus (Jesus’ nativity)
Moses (Moses’ leadership)
Greek (Hellenized) names: Demosthenes’ orations, Rameses’ tomb
Enceladus’

Note: When a proper noun is italicized, the possessive ending is roman.
the *Saturday Review*’s editor

Note: When the final “s” is silent, add “s”

Descartes's essays, Agassiz's lectures (typically a problem with French names)

3) Add the apostrophe to the **final element of a compound**.

someone's pen, someone else's proposal,
Clark and Watson's results

4) Add "s" to form the **possessive of a plural noun** not ending in "s"

children's, media's, deer's

5) Add an apostrophe alone (not followed by an "s") to form the **possessive of a plural noun ending in a sibilant**.

Eyewitnesses' account, scientists' opinions

6) Add an "'s" to form the plural of the following:

-symbols
* 's

In general, it is best to avoid making symbols plural. Use words instead.

Ex.: * 's should be changed to "asterisks"

-Words used as an example words
the which's and that's

-A single letter, especially when it would spell a word with "s" added
A's, a's, I's, i's, U's, u's, s's

-Abbreviations that retain periods or have subscripts or superscripts
 K_d 's [K_d values is preferred]

7) At *Science*, we rarely use contractions such as you're, he's, it's, didn't. Use "you are," "he is," "it is," "did not" except in quotes where contractions should be retained.

WHEN NOT TO USE AN APOSTROPHE

1) With a personal acronym or plural numbers
PVBs, threes, 12s, 1930s

2) With plural forms of academic degrees
BSs, MAs, PhDs

3) With a personal pronoun in the possessive case.
her, hers, their, theirs, its

4) With a possessive "s" in the first element of a well-established geographic name in the United

States (a British geographic name is likely to retain the apostrophe).

U.S.: Buzzards Bay Pikes Peak Woods Hole

British: Land's End (England) St. John's (Newfoundland)

5) In the name of an organization in which the qualifying term is treated as an adjective rather than as a possessive noun. If in doubt, check the name in reference source.

Authors League of America (but Showmen's League of America)

Teachers College (but Woman's Christian Temperance Union)

6) Do not add an apostrophe when forming the plural of acronyms, numbers, numerals, or years, except for those acronyms with subscripts or superscripts or that might be confused.

LDLs, twos, 7s, 1920s, RNAs, TCID₅₀'s, Ecp's (exterior cap, no Ecps)

7) With eponyms. These examples are *Science* style.

Wilms tumor

Down syndrome

Babinski sign

but: Alzheimer's disease

8) Items such as

"Wilson *et al.*'s results" should be changed to "results of Wilson *et al.*"

BRACKETS

USAGE

1) To enclose any matter, including bibliographic details, that either you or the author has inserted in a **quotation**.

"A major concern [for NSF] must be technology assessment."

"The average seeding effect in the entire region [comprising 100,000 km²] is a 21% loss of rain."

In the words of Sir William Lawrence Bragg, "The important thing in science is not so much to obtain new facts as to *discover new ways* [italics added] of thinking about them."

"Stanton reported that "heterogenous [sic] processes are responsible for this reaction."

2) To enclose **parenthetical material**.

We added a solution [5 M tris, bovine serum albumin (10 mg/ml), and Hcl]...

3) To indicate concentration: [Ca²⁺]

4) To enclose symbols representing absolute values.

5) Use italic letters within square brackets to form names for **polycyclic aromatic compounds**.

dicyclobuta[*de,ij*]naphthalene dibenz[*a,j*]anthracene
1H-benz[*de*]naphthalene indenol[1,2-*a*]indene

6) Use numbers separated by periods within square brackets to form names of **bridged and spiro alicyclic compounds**.

bicyclo[3.2.0]heptane spiro[4.5]decane

7) Use square brackets closed up to the compound name to indicate **isotopic labeling** or isotopic substitution.

[15N]alanine [2-14C]leucine
[15N]ammonia

8) In **math** expressions, such as $[d(\log kT)]P$

COLONS

USAGE

1) **Long quotations:** Use a colon to introduce a long quotation; use a comma for a short quotation.

2) **Lists:** Use a colon to introduce a list or enumeration not immediately preceded by a verb or preposition. The list could be made up of phrases or complete sentences.

for three reasons: (i) a clean surface under a vacuum (verb)...(*Science* **235**, 150)
questions requiring quantitative answers: When? How long ago? How often? (*Science* **235**, 157)

3) **Capitalization after a colon:** The first word after a colon usually should be capped when it begins a complete sentence that is an important statement by itself.

The scenario leaves researchers in something of a bind: Either their long-term prediction is about to fail or the prospects for short-term prediction will suffer a serious blow.

The first word after a colon does not have to be capped when the intent is to emphasize a close relation in thought between two independent clauses.

The effect of the ephemeral rise in water temperature on the degradation of polythionates would be small: the half-life of tetra- and pentathionates...(verb) (*Science* **235**, 1635)

4) **Ratios:** Use a colon to separate the parts of a ratio or to indicate the proportions of a mixture (no space before and after the colon). A slant line (solidus) is also acceptable.

ethanol and acetic acid (3:1)...in a 1:100 dilution of saturated...(Science **235**, 1624)

5) **Chromosomes:** Use a colon to symbolize breaks in human chromosomes (:, break, no reunion, as in a terminal deletion; : :, break and join).

46,XX,del (1)(pter arrow q21:)
46,XX,del (1)(pter arrow q21::q31 arrow qter)

Also use colons to indicate an insertion:

Ex: pJE201::Tn5 indicates transposon Tn5 is inserted into plasmid pJE201.

PLACEMENT OF THE COLON

Outside the closing quotation mark, parenthesis, or bracket.

WHEN NOT TO USE A COLON

1) Do not use a colon between a verb and its subject.

2) Do not use a colon after a title, heading, subheading, equation, or formula standing separate from the text.

COMMAS

USAGE

1) **Independent clauses:** Use a comma to separate independent clauses joined by a coordinating conjunction—such as and, or, nor, but, yet, for, so, and whereas—when they connect two complete thoughts.

Examples:

The individual layers were flat and uninterrupted across the width of the post, and they continued to within 5 Å of the surface. (*Science* **235**, 1629)

Many spectral lines are only one resolution element wide, so their cause must have persisted for 3 years. (*Science* **235**, 1632)

First, statistical associations between deposition rates and arsenic and cadmium trends are highly significant, whereas associations between the trends and the number of power plants, nonferrous smelters, and other industrial sources upstream of the stations are not

significant. (*Science* **235**, 1613)

Exceptions

– **Short clauses:** If both clauses are very short, the comma may be omitted if the omission does not result in ambiguity.

Ex: He is tall and she is short.

– **Semicolon with comma:** When a conjunctive adverb joins independent clauses, use a semicolon and comma or split into separate sentences.

Examples of conjunctive adverbs:

however, nevertheless, hence, moreover, accordingly, besides, also, thus, then, still, otherwise, furthermore, therefore, consequently

Ex: No physician was available; however, the patient recovered.

OR

No physician was available. However, the patient recovered.

– A semicolon may also be used between two complete clauses joined by a conjunction if the sentence is very complex or if one of the clauses is subdivided by commas.

Ex: Margaret, who had already decided that she would ask the question at the first opportunity, tried to catch the director's attention as he passed through the anteroom; but the noisy group of people accompanying the director prevented him from noticing her.

2) **Introductory clauses:** Use a comma to set off an introductory clause unless the clause is very short and can be read without a natural pause. Examples of words that begin such clauses: if, although, since, when, while, because, as, whereas, for.

Since 1959, no volcanic activity has been observed. (*Science* **235**, 1634)

Before evaluating Eq. 2 for our system of 13 theoretical frequencies, we will remove one of its 10 successful lines. (*Science* **235**, 1632)

After some years had passed, the situation changed. (*Words into Type*)

To illustrate the dramatic improvement, we will discuss three sets of diamond films. (*Science* **260**, 1310)

But: During the afternoon the plane landed in Uganda (*Chicago*)

Use a comma after introductory words to prevent misreading.

As the pond fills, the plants are killed off. (*Words into Type*)

In all, eight experiments were performed.

3) In a series of elements (words, phrases, or clauses) that contains three or more items, use commas before “and” and “or” Authors often drop the **serial comma**; be careful to insert.

The tomatoes wilted, the beans died in August, and the peppers bore no fruit.
...males of a *Drosophila stimulans* strain carrying the mutations *f*, *net pm*, and *st e* (forked bristles, *net* wing veins with *plum* eyes, and *scarlet* eyes with *ebony* body) on the X, 2, and 3 chromosomes. (*Science* **235**: 1636)
Thus, the unknown substances in the lake water could be hepta-, octa-, and nonathionates. (*Science* **235**: 1635)

– In a complex series (that is, if any of the elements contains internal punctuation), commas and semicolons may be used unless commas can be used by themselves without confusion.
CO₂, 40.443%; H₂S, 59%; and SO₂, 0.44% (*Science* **235**, 1634)

4) **Nonrestrictive clauses:** To separate a nonrestrictive clause or phrase, use a comma before and after as needed to separate it from the rest of the sentence. A nonrestrictive clause gives nonessential or descriptive information—for example, a clause properly after “which” or “who.” A good way to remember this is “that defines” and “which informs.”

The structure of these edge surfaces, which lies at the heart of the chemical properties of these materials, is poorly understood at present. (*Science* **235**, 1629)
The program is...meant to...demonstrate the feasibility of “new” technologies, which officials have yet to identify. (*Science* **235**, 1567)
Questions about the validity of Breuning’s work were first raised...by Robert Sprague of the University of Illinois, who had taken Breuning on as an investigator for an NIMH project... (*Science* **235**, 1566)

Sometimes the editor must change the author’s “which” to “that” to show that the phrase after it is restrictive (defining, limiting). Phrases that begin with “that” do not take commas.

...base-pair changes that create or destroy a cleavage site for a specific restriction enzyme... (*Science* **235**, 1617)
...regions of a tRNA molecules that are contacted by an amino acyl tRNA synthetase are not contiguous. (*Science* **235**, 1651)

There is another use of “which” that does not require commas. In such cases, there is no natural pause, and the information in such instances is not entirely descriptive or nonessential.

...the heating of a upper atmosphere by an initial burst would create different conditions for a subsequent burst that could produce new backgrounds against which sensors would have to operate. (*Science* **235**, 1605)

5) Use a comma, or commas, **to separate a nonrestrictive appositive** from the rest of the sentence. These appositives are those that are interchangeable with the item or items they are set off from. See section on appositives.

The volcano has three active craters, Mizugama, Yugama, and Karagama. The central

crater, Yugama, contains a lake. (*Science* **235**, 1633)
Periodic oscillations, called *r*-modes,... (*Science* **235**, 1631)
The mass density, *P*, is constant on... (*Science* **259**, 1868)
Raymond Turner, a mammologist, described the species.

But:

The fruit fly *Drosophila melanogaster* appears...
The mammologist Raymond Turner said that...
The Yugama crater contains a lake...

6) Use a comma or commas **to set off a conjunctive adverb** (see 1 above), transitional phrase (on the contrary, on the other hand, in fact, after all, in the first place), or parenthetical expression (that is, namely, for example, respectively).

Without a doubt, polythionates had been degraded (*Science* **235**, 1634)
However, a major fraction of peripheral blood lymphocytes.... (*Science* **260**, 1274)
...the highly conserved amino acids Leu, Cys, and Glu were changed to Arg, Arg, and His, respectively. (*Science* **260**, 1331)

If the break created by parenthetical expressions introduced by “that is” or “namely” is large, em dashes should be used instead of commas.

Identification of the parameters that control growth—for example, the nature of the deposited particles, the deposition rates, and substrate conditions—and their application in materials preparation techniques are of great basic and technological significance. (*Science* **260**, 1304)

7) **To introduce a short quotation** (less than three lines) or a direct question, use a comma—but do not use it if the quoted word or phrase is the subject or object of the sentence. (A colon is preferred before a long quotation.)

It is, says Mazur, “a very strange conjecture.” (*Science* **235**, 1572)
“Sixty-seven percent of our patients have infections,” says Wiley. (*Science* **235**, 1577)
...have data suggesting that the T4 receptor “is located on brain neurons.” (*Science* **235**, 1577)

8) **To separate multiple adjectives modifying the same noun** if they could be strung together by “and” without changing the meaning, use a comma.

Precision-made, highly reproducible surfaces (*Science* **235**, 1661)

9) **To show that words were left out**, use a comma. Usually, a comma is used for a verb that is omitted; the second example is a headline.

One aspect of the League might be entitled Cooperation in the Works of Peace; the other, Cooperation for the Prevention of War. (*Words into Type*)
U.S., Soviets Renew an Exchange (*Science* **235**, 1568)

10) **Contrasted expressions:** Use a comma to set off contrasted expressions or interdependent antithetical clauses. Strunk & White include “but” in this rule.

It is orange, not red.

11) **Words placed out of position for clarity:** Use a comma to set off words, phrases, or clauses placed out of their natural position for emphasis and clarity.

Keys provide, except in the most specialized works, a useful means of identification.

12) Use a comma in a **five-digit number**. The only exceptions to this are five-digit page numbers, numbers in addresses, or years. (See the numbers section for more details.)

The herd included 15,600 adults.

J. Biol. Chem. **263**, 15675 (1988)

13) Use a comma **to separate adjacent sets of numbers**.

In 1935, 100 experiments were completed.

14) Bibliographic references: Use a comma to separate certain elements in bibliographic references.

In text: (10, 11)

but: (10–15) (en dash)

author *et al.* (no comma)

In the reference section:

J.C. Smith Jr., H.C. Morse III, D. Johnson...

J. Smith Jr. *et al.*,...

author, author, (2 authors)

author *et al.*,

author, author, author

Book Title, editor's name, Ed. (publisher, city of publication, ed. 2, 1968)

Journal name **volume no.**, page no. (year)

15) Use commas in **addresses** as follows, but not before or after “Jr.”:

412 Fifth Avenue, Newark, DE 11111, USA.

Donald Brown Jr. is the guest of honor

J.C. Smith Jr., D. Brown, S. Smith, Department of Biology,...

16) **Chromosome aberrations:** To separate some elements in symbolic descriptions of human chromosome aberrations, use a comma without a space.

46,XX,t(4;13) (p21;q32)

17) **Chemical names:** Use a comma between locants without a space (see *ACS Style Guide*).

1,2-dicyanobutane; beta,4-dichlorocyclohexanepropionic acid

but: 17beta-hydroxy-5alpha-pregane

18) **Amino acids:** To separate amino acid symbols in amino acid residues of unknown sequence, use a comma without a space.

Gly,Phe,Tyr

WHEN NOT TO USE A COMMA

1) Do not use a single comma to separate a verb from its subject or object.

(An intervening phrase may require two commas between subject and verb.)

Incorrect: The solvents used in this study were, cyclohexane, methanol, *n*-pentane, and toluene.

Correct: The solvents used in this study were cyclohexane, methanol, *n*-pentane, and toluene. (ACS, 18)

2) Do not use a comma to separate the parts of a compound predicate that is not a series.

Incorrect: The data were obtained, and are listed in Table 1.

Correct: The data were obtained and are listed in Table 1.

3) Do not place a comma before or after a restrictive (defining) appositive.

The species *Bombyx mori* was described later.

4) Do not place a comma between digits in page number, an address, or a four-digit year number.

p. 6984 5606 Glenwood Road 1066 CE 3000 BCE 35,000 BCE
[Not AD and BC, but CE (of the common era) or BCE (before the common era)]

5) Do not use a comma after a title, heading, subheading, equation, or formula standing separate from the text. [example omitted]

6) Do not use a comma to splice together two complete sentences.

Wrong: Long-term survival could be pure luck, nevertheless, I would expect genetic, behavioral, and environmental factors to influence the rate of progression to AIDS.

Correct: Long-term survival could be pure luck. Nevertheless, I would expect... (*Science* 260, 1274)

Or: Long-term survival could be pure luck; nevertheless, I would expect ...

AMBIGUITIES OF USAGE

When used properly, commas indicated interruptions in thought or sentence structure and thus should help the reader. But even when the rules above are adhered to, it may be a matter of judgement on the part of the author (and editor) whether to use a comma.

Ex: On Tuesday it rained and the humidity increased.

In this example, a comma after “Tuesday” would imply that both events happened on that day. A

comma after “rained” could imply that the two events were unrelated. Using both commas would seem to be proper from the rules above but that would make the sentence unnecessarily jerky:

On Tuesday, it rained, and the humidity increased.

Similarly, clauses beginning with “because” that do not begin a sentence require a judgment call. Usually, if the clause is restrictive, no comma is required.

Millions of stars cannot be seen with the eye because they are too far away to supply the light necessary for vision. (*Words into Type*)

For more examples, see *Words into Type* and *Chicago Manual of Style* on the use of commas.

PLACEMENT OF COMMAS

- Place the comma inside the closing quotation mark when a sentence continues beyond the end of the quoted matter, even when the comma is not part of the quotation.

He said, “I hate dogs,” even though it’s a lie.

- After a period that follows and abbreviation if the sentence requires a comma.

In 3000 B.C., these stones...

- Outside a closing parenthesis or bracket, never before an opening parenthesis or bracket.

...at the river’s edge), but...

- Mark to close up if the comma is not followed by a space (as in amino acid residues of unknown sequence).

adenosine 3',5'-monophosphate

$f(x,y),(x_i,y_i)$ (coefficients, functional coefficients)

$(\text{Mg,Fe})_2\text{SiO}_4$

$F(2,40)$ (degrees of freedom in statistics)

Na^+,K^+ -ATPase

See also 16) chromosomal aberrations and 17) chemical names (above)

ELLIPSIS MARKS

USAGE

1) Use the ellipsis to indicate **omission of a word or group of words** within quoted material. In such instances, it is best not to retain punctuation marks (commas, semicolons, etc.) around the ellipsis.

“Laboratory findings . . . were similar in three patients.”

2) To indicate **omission of words from the end of a sentence** within quoted material, place the period or other punctuation before the ellipsis.

“Platelet serotonin release is only one manifestation. . . .”

“Why is it that they array themselves against me? . . . Where were they during the rebellion?” (*Chicago*)

WHEN NOT TO USE ELLIPSIS MARKS

1) Do not use the ellipsis if it is implied by use of quotation marks that words have been omitted.
Smith characterized the outbreak as “the heaviest infestation” he had seen in 1975.

2) Do not use the ellipsis if the context indicates that some of the words have been omitted.
Morgan stated that a science writer “must consider whether his meaning will be clear to students of literature.”

Use of the lowercase m in must in the example above implies that the first part of the sentence has been omitted. If “to students of literature” were omitted, the sentence should end thus: will be clear”

EXCLAMATION POINT

USAGE

The exclamation point is rarely justified in scientific writing except as a factorial symbol in mathematics. The editor removing an exclamation point may have to rewrite the sentence so that it no longer requires one.

Ex:use scientific technology. What an accomplishment that would be!

Can be rewritten as:

...use scientific technology, which would be an important advance.

PARENTHESES

USAGE

1) Use the parentheses to enclose a **comment or explanation** that is structurally independent of the sentence.

The total amount (10 mg) was recovered by modifying the procedure.

The tyrosol radical was first believed to be directly involved in the abstraction of the hydrogen at C-3 (Fig. 2). (*Science* **260**, 1774)

DOE is not using the best available technology (ultrasonics) to locate cracks. (*Science* **235**, 1563)

2) Use parentheses to group **mathematical expressions**

$$(a + x)/b$$

3) Use parentheses to label **enumerations within a paragraph**.

A number of techniques would be available for this task: (i) partitioning the overall job into well-defined parts with rigorously determined interfaces, (ii) using already proven software routines when available, (iii) using common... (*Science* **235**, 1605)

- Reference may be made to one of the series by citing it in parentheses the partitioning mentioned in (i) indicates that...

Exception:

- To label enumerations of paragraphs, use a single closing parenthesis.
 - 1) The paragraph begins and continues until it ends on the next line, in this case.
 - 2) The next paragraph, longer perhaps, goes on.

4) Use parentheses for **text citations of references** (italic numbers or date and page) and for **numbers of displayed equations**.

into the host genome (16–18).

(*Science*, 2 January, p. 17) (Note: this example and the one below are used for news stories, and the year is not needed if the reference is to an article in the current year.)

(*Science*, 19 December 1986, p. 1488)

$(a + x)/b$ (3) [equation]

Note: In the text of *Science* we refer to a reference as (12) but to Eq. 3 (no parentheses).

5) Use parentheses for labeled **parts of a figure**.

(A)

(A and C) or

(A) and (C)

6) Use parentheses to enclose an abbreviation or acronym that follows the written-out term:
antigen (Ag)

7) Use parentheses to enclose a **trademark or manufacturer** of reagents or equipment.
cobalt chloride (Pharmacia, Piscataway, NJ)

SPECIALIZED USES FOR PARENTHESES

1) Use parentheses to enclose the name of the author of the original taxonomic description when a species is transferred to a genus other than the one to which it was originally assigned.

Shigella dysenteriae (Shiga) Castellani and Chalmers, which was first identified by Shiga as *Bacillus dysenteriae*.

2) Use parentheses with immunoglobulins and genetic factors of Gmv and Inv. See *CBE*

- 3) Use parentheses to enclose chromosome numbers in symbolic representations of animal chromosome aberrations. See *CBE* and Commas section of style guide.
- 4) Use parentheses to enclose elements in organic chemical names that must be followed by subscript numbers applying to the entire radical, for example,

$$\text{CH}_3(\text{CH}_2)_{10}\text{CH}_2\text{OH}$$
- 5) Enclose oxidation numbers in parentheses, closed up to the element name or symbol, in text but not in formulas. (In formulas use superscripts.)
 iron(III) or Fe(III)
- 6) After a congressman's name to indicate party affiliation and state:
 John Glenn (D–OH) (en dash)
- 7) In nuclear reactions:

$$^{27}\text{Al}(\text{p},\text{d})^{26}\text{Si}$$

PUNCTUATION WITH PARENTHESES

- 1) Place the period outside the parentheses when the parenthetical material is not a full sentence.
 Proteins phosphorylated on tyrosine were detected by protein immunoblotting with antibodies to phosphotyrosine (anti-phosphotyrosine). (*Science* **260**, 1806)
- 2) In full sentences, the period goes inside the parentheses.
 Ex:
 ...efficient implementation. (Spectral methods also use complex and costly communications patterns, which poses additional challenges to implementation.)
- 3) Do not use parentheses within parentheses except for a scientific name and its authority that are in parenthetical matter.
 (*Fryxellia pygmae* (Correll) Bates)

PERIODS

USAGE

- 1) **Declarative sentences:** Use a period after a declarative or imperative sentence.
 The first projects would be selected in April 1988. (*Science* **235**, 1567)
- 2) **Abbreviations:** See the Acronyms and Units sections for abbreviations with and without periods.
 With periods: S. A. Forbes Jr., no., *et al.*; (time) p.m.

Without periods:

two-letter abbreviations for states: WV, ME [Use only in affiliations, addresses, and parentheticals that denote a company location; e.g., “(catalog no. 156847, Genentech, San Francisco, CA)”]; spell out in running text. But never spell out “DC” (as in “Washington, DC).”]

country abbreviations: USA, UK (in author affiliations and addresses); US, UK (as nouns or adjectives in running text; but note that “United States” and “United Kingdom” may also be spelled out as nouns)

organizations: NIH, WHO

directions: N, SE

other examples: NDA, RBC, log, exp

academic degrees: BS, MA, PhD

PLACEMENT OF PERIODS

1) Inside the closing quotation mark in a sentence ending with a quoted word, even though the period is not part of the quotation.

Only much larger areas, he says, can preserve what he calls “wildness.” (*Science* **260**, 1870)

2) Inside the closing parenthesis or bracket when the parenthetical matter is an independent sentence.

(For comparison, the Nagasaki bomb contained 6 kg of plutonium.) (*Science* **235**, 1555)

3) Outside the closing parenthesis or bracket when the parenthetical matter is a dependent element at the end of the sentence.

more than 300,000 kilograms of plutonium (20) (Fig. 1). (*Science* **235**, 1555)

divide between increases and decreases (for example, total phosphorus and suspended sediment). (*Science* **235**, 1609)

4) Before ellipsis marks when the omitted material is at the end of the sentence (also see *Chicago*,)

“at a higher power level than may have been safe. . . .” (*Science* **235**, 1563)

WHEN NOT TO USE A PERIOD

1) In or after certain abbreviations (see Acronyms and Units sections).

2) After a period that follows an abbreviations at the end of a sentence.

3) After an entry in a table (unless it is an abbreviations that requires a period).

4) After an entry in a list that is not a complete sentence. Usually, another form of punctuation is appropriate.

Ex: The myosin primary binding site is potentially composed of three types of interactions with actin: (i) an ionic interaction involving a flexible loop, (ii) a stereospecific interaction, and (iii) a strengthening of this interaction. (*Science* **261**, 62)

5) After a title, heading, equations, or formula standing separate from the text.

QUESTION MARKS

USAGE

– Use a question mark **after a direct question**.

How is the resistance of a wire affected by doubling its length? by halving its diameter?

In a series of questions, use only one question mark at the end when the form is like the following:

What is meant by the terms atom, molecule of an element, and a molecule of a compound?

Make sure to place the question mark only at the end of the sentence or clause that contains the question, not elsewhere:

Were the molecules real? the scientists wondered.

Whether a question should begin with a capital letter when it follows an introductory element is for the editor to decide. Generally, the more formal the question, the more usual it is to begin with a capital letter.

The question arises, What is the role of government in medical care?

Before deciding, ask yourself, will it work?

– Use a question mark carefully **with other forms of punctuation**. See *Words into Type*, where these rules are outlined.

with a period: if an interrogative sentence and a declarative sentence end at the same time, retain the question mark and omit the period.

with a comma: if a question ends within a sentence and the construction calls for a comma at the same point, the comma should in most cases be omitted and the question mark retained.

with a semicolon: when the sentence calls for a semicolon where a question ends, the semicolon is usually retained along with the question mark.

with quotation marks and parentheses or brackets: when used in conjunction with closing quotes or parentheses, the question mark should precede the quotation marks or

parentheses if it is part of the quoted matter; otherwise, it should follow.

– Where a question mark is **part of the title of a work**, retain the question mark along with any other punctuation (except terminal periods). If the question mark in such cases is at the end of the sentence, retain it and do not use any terminal punctuation (period). See *Words into Type*. In most cases, a sentence can be rearranged so that awkward doubling of punctuation will not be necessary.

– Use a question mark to express **doubt about a date**. This is very infrequently used.
Edmund Spenser (1552?–1599)

WHEN NOT TO USE A QUESTION MARK

– With indirect questions.

Early theorists wondered why Earth revolved around the Sun.

– In requests that are “disguised” as questions

Will you please rush the manuscript to the printer.

– When the question consists of one word. Quotation marks may or may not be used in such cases, but the words should never be italic and should not be capitalized.

The question was no longer “how” but “when.”

QUOTATION MARKS

USAGE

The rules given below are those of U.S. usage. Many British authors and editors use single quotation marks to enclose primary quotations and double quotation marks to enclose secondary quotations. Be careful to check and use according to the rules below.

DOUBLE QUOTATION MARKS

- 1) For a direct, primary quotation (never indirect quotations).
- 2) To set off a tertiary quotation (a quotation within a quotation within a quotation).
- 3) For the title of an article, the title of a chapter or other part of a book, or the title of a series, except when the title is in a bibliographic list.
- 4) For a new technical term, or an old term applied in a new or unusual sense. *Science* style is to enclose the new term in quotes the first time it is used but not thereafter. For old terms applied in an unusual sense, quotes may be used each time the term is used.

SINGLE QUOTATION MARKS

- 1) To enclose a secondary quotation (quoted matter within a quotation).
“An obvious explanation is that the detection techniques are not sensitive enough. But one of the questions that people have raised is, ‘Are there some indirect methods by which the virus damages the nervous system?’” (*Science* **235**, 1576)
- 2) For the names of plant cultivars.
Syringas vulgaris ‘Mont Blanc’
Chamaecyparis lawsoniana cv. ‘Silver Queen’

PLACEMENT OF QUOTATION MARKS

- 1) At the beginning of each paragraph of quoted text; if the quotation is more than one paragraph, close the quotation by a quotation mark at the end of the last paragraph only. (Long quotations may be set off from the text with space above and below, in which case quotation marks are not needed.)
- 2) Outside of a comma or a period, even if it is not part of the quotation.
- 3) Inside of a semicolon or colon.
- 4) Outside of any other punctuation mark if it is part of the quotation; inside if it is not.

SEMICOLONS

USAGE

- 1) Use a semicolon **to separate coordinate independent clauses** not joined by a conjunction.
First, we get an early flight demonstration of TOS; if we delay until 1992, that’s another 26 months. (*Science* **235**, 1568)
- 2) Use a semicolon **between independent clauses** joined by a conjunctive adverb.
We worked until the tide turned; however, we found too few samples.
- 3) Use a semicolon **to separate the parts of a complex series**, that is, elements containing internal punctuation (any one of them may contain a comma).
The compounds studied were methyl ethyl ketone; sodium benzoate; and acetic, benzoic, and cinnamic acids.
- 4) Use a semicolon **to separate chromosome numbers** in symbolic designations of chromosomal aberrations.
46,XX,t(4:13) (p21;q32)

5) Use a semicolon **to separate symbols for mutant** genes on different chromosomes of the fruit fly.

bw; e; ey

PLACEMENT OF SEMICOLONS

Outside a closing quotation mark, parenthesis, or bracket.

WHEN NOT TO USE A SEMICOLON

- 1) Do not use a semicolon between dependent and independent clauses.
- 2) Do not use a semicolon after a title, heading, subheading, equation, or formula standing separate from the text.